

Data Warehousing Assignment 2

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1. TASK1 – Data Integration (ETL) and Maintenance

1.1 Original datasets

Answer:

1. Workspace Name: c7263731
2. User Name: c7263731
3. Password : Manreet1

Creating a New Schema (Name Data warehouse)

Data Set Loading in APEX:

The screenshot displays the Oracle APEX Data Workshop interface. The main content area features a 'Get Started' section with two buttons: 'Load Data' (Load CSV, XLSX, XML, JSON data) and 'Unload Data' (Unload data in text or XML format). Below this is a table titled 'Recently Created Tables'.

Table Name	Created
CRIMESS	2 days ago
HTMldb_PLAN_TABLE	2 days ago
DRUGS	2 days ago
CRIME_BURGLRY	2 days ago
CRIME_FACT	5 weeks ago
TIME_DIM	5 weeks ago
STATION_DIM	5 weeks ago

The right sidebar contains three sections: 'About' (Use Data Workshop page to load data from various file formats...), 'Schema' (Select the default database schema for your Data Workshop session. C7263731), and 'Tasks' (Load to SODA Collection).

Data Loaded in APEX

The screenshot shows the 'Load Data' dialog in APEX. The 'Copy and Paste' tab is selected. A message states: 'You can copy and paste CSV data, or select a sample data set. - Select Sample -'. Below this, a table of data is displayed. The table has 12 columns: Month, Force, Neighbourhood, All crime, Anti-social behaviour, Burglary, Criminal damage and arson, Drugs, Other theft, Public disorder and weapons, Robbery, and Shoplifting. The data is organized by month and force, with specific counts for each category.

Month	Force	Neighbourhood	All crime	Anti-social behaviour	Burglary	Criminal damage and arson	Drugs	Other theft	Public disorder and weapons	Robbery	Shoplifting			
2019-03	West Yorkshire Police	AAN_HP	290	91	27	33	11	64	8	6	3	14	27	6
2019-03	West Yorkshire Police	BA_BRH	307	90	24	48	11	38	7	9	14	28	31	7
2019-03	West Yorkshire Police	CAN_CITY	1032	231	44	39	25	365	27	3	114	25	103	46
2019-03	West Yorkshire Police	CAN_CH	643	224	0	67	14	53	21	7	39	78	75	12
2019-02	West Yorkshire Police	AAN_HP	251	75	18	28	9	63	7	12	2	6	26	5
2019-02	West Yorkshire Police	BA_BRH	348	99	31	36	13	30	7	7	10	69	38	8
2019-02	West Yorkshire Police	CAN_CITY	1088	259	53	33	16	482	28	5	149	37	80	26
2019-02	West Yorkshire Police	CAN_CH	667	227	68	95	13	80	14	5	28	58	68	19
2018-12	West Yorkshire Police	AAN_HP	282	57	17	15	9	44	3	5	6	11	27	8
2018-12	West Yorkshire Police	BA_BRH	348	84	31	39	37	31	4	13	7	54	42	6
2018-12	West Yorkshire Police	CAN_CH	628	193	46	80	22	53	22	8	22	71	84	25
2018-12	West Yorkshire Police	CAN_CITY	1227	254	29	40	33	459	26	9	144	51	142	31
2018-11	West Yorkshire Police	AAN_HP	331	109	49	29	17	69	4	7	6	12	23	6
2018-11	West Yorkshire Police	BA_BRH	379	112	54	43	16	35	5	5	8	50	40	11
2018-11	West Yorkshire Police	CAN_CH	648	193	82	89	14	65	9	10	25	69	69	23
2018-11	West Yorkshire Police	CAN_CITY	1164	232	30	44	24	456	37	7	143	32	136	23
2018-10	West Yorkshire Police	AAN_HP	336	112	50	24	20	60	9	14	2	14	24	7
2018-10	West Yorkshire Police	BA_BRH	392	125	35	57	12	0	5	10	8	60	29	7
2018-10	West Yorkshire Police	CAN_CH	744	255	51	73	24	80	29	6	24	84	94	24
2018-10	West Yorkshire Police	CAN_CITY	290	91	27	33	11	64	8	6	3	14	27	6

Data Extraction Creating Table

The screenshot shows the 'Load Data' dialog in APEX. The 'Pasted Data' tab is selected. The 'Where do you want to load this data?' section is visible. The 'Load To' dropdown is set to 'New Table'. The 'Table Owner' is 'CT263731'. The 'Table Name' is 'CRIME_TABLE'. The 'Primary Keys' are 'SYS_GUID' and 'Identity Column'. The 'Use Column Data Types' checkbox is checked. The 'Settings' section shows 'Column Headers' set to 'First line contains headers', 'Column Delimiter' set to 'tab', 'Enclosed By' set to 'None', and 'File Encoding' set to 'Unicode UTF-8'. The 'Preview' section shows a message: 'Parsed first 42 rows to sample the column types. The preview below only displays the first 10 columns and 5 rows. To view the full preview and configure data load settings, click the Preview button.' The 'Load Data' button is highlighted.

Dataset Loaded Successfully

The screenshot shows the 'Load Data' dialog in Oracle APEX. The status message indicates: 'Table CRIME_TABLE created! 41 rows loaded successfully. 1 rows rejected.' Below this, a section titled 'Rejected Rows in CRIME_TABLE_ERR\$ Table' provides details on the rejected data. A table lists the first 10 rejected rows, including an error message and various attributes.

ERROR_MESSAGE	COL001	COL002	COL003	COL004	COL005	COL006	COL007	COL008	COL009	COL010
ORA-01722: invalid number	Month	Force	Neighbourhood	All crime	Anti-social behaviour	Burglary	Criminal damage and arson	Drugs	Other theft	Public disorder and weapons

1. SCREEN SHOT (new database schema)

The screenshot shows the 'Request Database Schema' dialog in Oracle APEX. The 'Schema' section has two radio buttons: 'Request a new schema' (selected) and 'Use an existing schema'. The 'Schema Name' field contains the text 'Data_Warehouse'. The 'Next' button is visible at the bottom right.

Creating a table (DRUGS)

The screenshot shows the APEX SQL Workshop interface. The SQL Commands pane contains the following code:

```
1 CREATE TABLE drugs (  
2   drugs_id number (1) Primary key,  
3   drugs_total number (20),  
4   force_varchar (20),  
5   neighbour varchar(20)  
6 );  
7
```

The Results pane shows the message "Table created." and the execution time "0.01 seconds".

Table Data entries & data type view

The screenshot shows the APEX SQL Workshop interface with the SQL Commands pane containing the following code:

```
1 select * from drugs  
2 describe drugs;
```

The Results pane shows the 'Describe' view of the 'drugs' table. The table structure is as follows:

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
DRUGS	DRUGS_ID	NUMBER	-	1	0	1	-	-	-
	DRUGS_TOTAL	NUMBER	-	20	0	-	✓	-	-
	FORCE_	VARCHAR2	20	-	-	-	✓	-	-
	NEIGHBOUR	VARCHAR2	20	-	-	-	✓	-	-

2nd DB CRIME TYPE

Table Finder - Google Chrome

Not secure | https://aet-oracle5.aet.leedsbeckett.ac.uk:8443/ords/f?p=4500:24:1969108602322::SC::P24_TABLE_NAME:CRIME_BURLRY

Search: Views: ☐ Rows: 15 Go

Close

Tables Columns

Table: CRIME_BURLRY

Column	Data Type	Length	Precision	Scale	Nullable
BURLR_ID	NUMBER	22	1	0	No
BURGALRY_TOTAL	NUMBER	22	20	0	Yes
FORCE_	VARCHAR2	20	-	-	Yes
NEIGHBOUR	VARCHAR2	20	-	-	Yes

1 - 4

```
select  
BURLR_ID,  
BURGALRY_TOTAL,  
FORCE_ ,  
NEIGHBOUR  
from CRIME_BURLRY
```

Append Replace

Search the web and Windows

Address: 1:52 PM 1/3/2022

SQL Workshop Commands History

SQL Workshop

Not secure | <https://aet-oracle5.aet.leedsbeckett.ac.uk:8443/ords/f?p=4500:3002:1969108602322::>

Object Browser SQL Commands SQL Scripts Utilities RESTful Services

Schema

Select the default database schema for your SQL Workshop session.

C7263731 Set

Create Object

- Table >
- View >
- Index >
- Sequence >
- Type >
- Package >
- Procedure >
- Function >
- Trigger >
- Database Link >
- Materialized View >
- Synonym >
- SODA Collection >

Recently Created Tables

HTMLDB_PLAN_TABLE	7 minutes ago
DRUGS	22 minutes ago
CRIME_BURLRY	23 minutes ago
STATION_DIM	5 weeks ago
TIME_DIM	5 weeks ago
CRIME_TYPE_DIM	5 weeks ago
CRIME_FACT	5 weeks ago
PL_REPORTED_CRIME	2 months ago
PL_WORK_ALLOCATION	2 months ago
PL_STATEMENT	2 months ago
PL_AREA	2 months ago
PL_STATION	2 months ago
PL_POLICE_EMPLOYEE	2 months ago
PL CRIME TYPE	2 months ago

Recent SQL Commands

select * from burglary;	3 minutes ago
select * from burglary;	3 minutes ago
select * from drugs describe burglr...	4 minutes ago
select * from drugs desc * drugs	5 minutes ago
select * from drugs desc * drugs	6 minutes ago
select * from drugs desc drugs	6 minutes ago
select * from drugs	7 minutes ago
select * from drugs	7 minutes ago
INSERT INTO drugs (drugs_id, total...	11 minutes ago
INSERT INTO drugs (drugs_id, durgs...	11 minutes ago
INSERT INTO drugs (drugs_id, durgs...	14 minutes ago
INSERT INTO drugs (drugs_id, durgs...	15 minutes ago
INSERT INTO drugs (drugs_id, durgs...	16 minutes ago
CREATE TABLE drugs (drugs_id n...	22 minutes ago

Recent SQL Scripts

DW assignment1_ss	5 weeks ago
DW assignment1_ss drop	5 weeks ago
prcs1	2 months ago
prcs	2 months ago

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Application Express 211.2

Search the web and Windows

Address: 1:54 PM 1/3/2022

Evidence Code

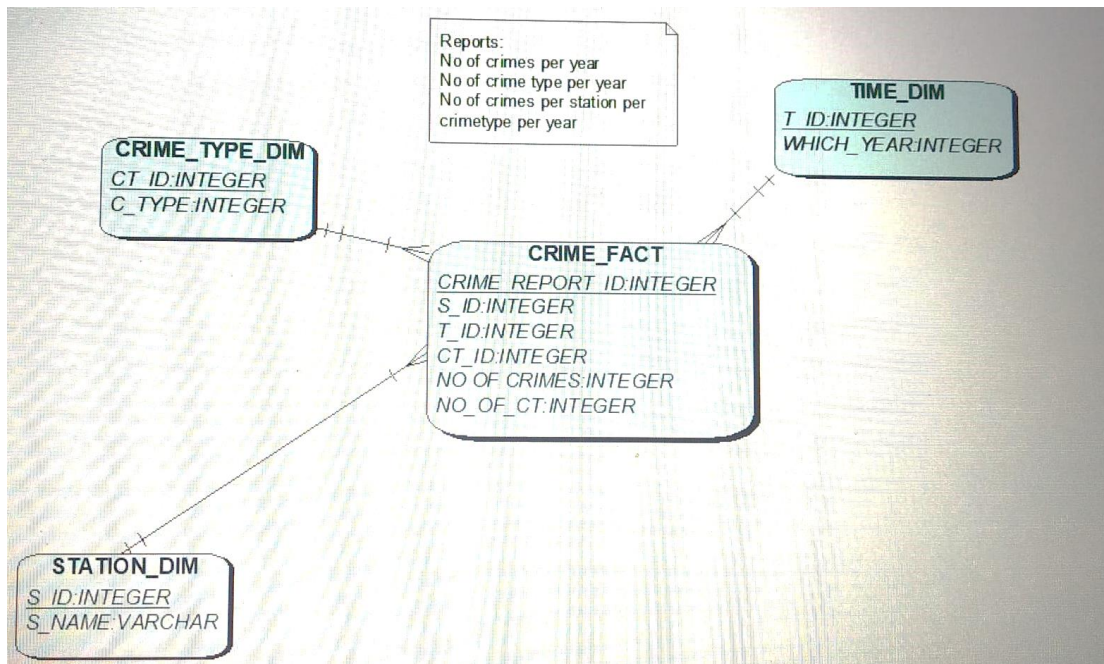
The screenshot shows the Oracle APEX SQL Workshop interface. The 'SQL Commands' tab is active, displaying the following SQL code:

```
1 CREATE TABLE drugs (  
2   drugs_id number (1) Primary key,  
3   drugs_total number (20),  
4   force_varchar (20),  
5   neighbour varchar(20)  
6 );  
7
```

The 'Results' tab shows the message 'Table created.' and the execution time '0.01 seconds'. The interface includes a top navigation bar with 'APEX', 'App Builder', 'SQL Workshop', 'Team Development', and 'Gallery'. The bottom status bar shows the user 'c7263731', the schema 'c7263731', and the application 'Application Express 2112'.

1.2 Star Schema

- Have the diagram of your SS here.



Fact Table

The screenshot shows the Oracle APEX Object Browser interface. The left sidebar lists various database objects, with **CRIME_FACT** selected. The main panel displays the table structure for **CRIME_FACT** in the **C7263731** schema. The table has five columns: **CRIME_REPORT_ID** (NUMBER, Primary Key), **FK1_S_ID** (NUMBER), **FK2_CT_ID** (NUMBER), **FK3_T_ID** (NUMBER), and **NO_OF_CT** (NUMBER). The **NO_OF_CT** column is nullable, while the others are not. The interface includes tabs for Data, Indexes, Model, Constraints, Grants, Statistics, UI Defaults, Triggers, Dependencies, SQL, REST, and Sample Queries. The bottom status bar shows the application is running on Oracle Express 211.2.

Table	Data	Indexes	Model	Constraints	Grants	Statistics	UI Defaults	Triggers	Dependencies	SQL	REST	Sample Queries
CRIME_FACT												
Column Name	Data Type	Nullable	Default	Primary Key								
CRIME_REPORT_ID	NUMBER	No	-	1								
FK1_S_ID	NUMBER	No	-	-								
FK2_CT_ID	NUMBER	No	-	-								
FK3_T_ID	NUMBER	No	-	-								
NO_OF_CT	NUMBER	Yes	-	-								

Crime register

The screenshot shows the Oracle APEX Object Browser interface. The left sidebar lists various database objects, with **CRIME_REGISTER** selected. The main panel displays the table structure for **CRIME_REGISTER** in the **C7263731** schema. The table has ten columns: **CRIME_ID** (NUMBER(4,0), Primary Key), **CRIME_NAME** (VARCHAR2(20)), **LOCATION_ID** (NUMBER(8,0)), **REPORTED_DATE** (DATE), **POLICE_ID** (NUMBER(4,0)), **CRIME_STATUS** (VARCHAR2(8)), **CLOSED_DATE** (DATE), **REPORTER_ID** (NUMBER(8,0)), **OFFICER_NOTE** (VARCHAR2(128)), and **CRIME_TYPE** (VARCHAR2(15)). The **CRIME_ID** column is the primary key. The **CRIME_NAME** and **CRIME_STATUS** columns are nullable, while the others are not. The interface includes tabs for Data, Indexes, Model, Constraints, Grants, Statistics, UI Defaults, Triggers, Dependencies, SQL, REST, and Sample Queries. The bottom status bar shows the application is running on Oracle Express 211.2.

Table	Data	Indexes	Model	Constraints	Grants	Statistics	UI Defaults	Triggers	Dependencies	SQL	REST	Sample Queries
CRIME_REGISTER												
Column Name	Data Type	Nullable	Default	Primary Key								
CRIME_ID	NUMBER(4,0)	No	-	1								
CRIME_NAME	VARCHAR2(20)	Yes	-	-								
LOCATION_ID	NUMBER(8,0)	Yes	-	-								
REPORTED_DATE	DATE	No	-	-								
POLICE_ID	NUMBER(4,0)	Yes	-	-								
CRIME_STATUS	VARCHAR2(8)	Yes	-	-								
CLOSED_DATE	DATE	Yes	NULL	-								
REPORTER_ID	NUMBER(8,0)	Yes	-	-								
OFFICER_NOTE	VARCHAR2(128)	Yes	-	-								
CRIME_TYPE	VARCHAR2(15)	Yes	-	-								

Crime reporting table

The screenshot shows the APEX SQL Workshop interface. On the left, a list of tables is visible, including CRIME_BURGLARY, CRIME_FACT, CRIME_REGISTER, CRIME_REPORTER, CRIME_REPORT_VIEW, CRIME_STAT, CRIME_TYPE_DIM, DRUGS, FINANCIAL, HTMLDB_PLAN_TABLE, LOCATION, OFFENCE, OFFICER, PL_AREA, PL_CRIME_TYPE, PL_POLICE_EMPLOYEE, PL_REPORTED_CRIME, PL_STATEMENT, and PL_STATION. The main area displays the structure of the CRIME_REGISTER table with columns: CRIME_ID, CRIME_NAME, LOCATION_ID, REPORTED_DATE, and CRIME_STATUS. Below the structure, the table's data is shown in a grid format.

CRIME_ID	CRIME_NAME	LOCATION_ID	REPORTED_DATE	CRIME_STATUS
1	Theft	1	05/24/2015	CLOSED
2	hit and run	1	11/02/2015	OPEN
3	armed robbery	2	01/15/2016	OPEN
4	murder	4	10/29/2017	CLOSED
5	armed robbery	1	02/13/2019	OPEN
6	Black mail	2	05/09/2019	CLOSED

Aggregate Function in APEX

1. Count()
2. Sum()
3. AVG()
4. Max()
5. MIN()

First COUNT () is applied on crime register table as follows:

The screenshot shows the APEX SQL Workshop interface with the SQL Commands tab selected. The query editor contains the following SQL code:

```
1 select count(*)
2 from crime_register;
```

The Results tab is active, showing the output of the query. The result is a single row with the value 14 under the column heading COUNT(*). Below the result, it states "1 rows returned in 0.00 seconds".

COUNT(*)
14

SUM()

The screenshot shows the APEX SQL Workshop interface. The SQL command area contains the following code:

```
1 select count(*)
2 from crime_register;
3
4 select sum(location_id)
5 from crime_register;
```

The Results tab is selected, showing a single row with the value 32 under the column heading SUM(LOCATION_ID). The status bar indicates "1 rows returned in 0.01 seconds".

SUM(LOCATION_ID)
32

AVG()

The screenshot shows the APEX SQL Workshop interface. The SQL command area contains the following code:

```
1 select count(*)
2 from crime_register;
3
4 select sum(location_id)
5 from crime_register;
6
7 select avg(location_id)
8 from crime_register;
```

The Results tab is selected, showing a single row with the value 2.28571428571428571428571428571429 under the column heading AVG(LOCATION_ID). The status bar indicates "1 rows returned in 0.00 seconds".

AVG(LOCATION_ID)
2.28571428571428571428571428571429

MAX()

The screenshot shows the APEX SQL Workshop interface. The SQL command area contains the following code:

```
4 select sum(location_id)
5 from crime_register;
6
7 select avg(location_id)
8 from crime_register;
9
10
11 select max(location_id)
12 from crime_register;
```

The Results tab is selected, showing a table with the title **MAX(LOCATION_ID)** and one row containing the value 5. Below the table, it states "1 rows returned in 0.00 seconds" and provides a "Download" link.

MIN()

The screenshot shows the APEX SQL Workshop interface. The SQL command area contains the following code:

```
8 from crime_register;
9
10
11 select max(location_id)
12 from crime_register;
13
14
15 select min(location_id)
16 from crime_register;
```

The Results tab is selected, showing a table with the title **MIN(LOCATION_ID)** and one row containing the value 1. Below the table, it states "1 rows returned in 0.00 seconds" and provides a "Download" link.

Answer: star schema based on fact table, mostly the fact tables stores the foreign keys of other associated table in it. All associated table has been directly implemented and associated with fact table.

in assignment the data dictionary table prepared in excel sheet which present the data types and data size of the attribute of the table.

Star schema created in excel as well which present the fact table and other associated table , related with fact table entries.

```
Select count(*)  
from crime_register;
```

```
Select sum(location_id)  
from crime_register;  
select avg(location_id)  
from crime_register;
```

```
select max(location_id)  
from crime_register;  
select min(location_id)  
from crime_register;
```

1.3 Staging Area

Answer: in this project I am considering Excel sheet instead of using tableau, excel worksheet presented well in order to create star schema and other related table with data dictionary.

Data Dictionary Table

Data Dictionary Table			
Field Name	Caption	Data type	Field Size
str_Month	Month	Date	10
str_force	Force	Text/Varchar	100
str_Neighbour	Neighbourhood	Text/Varchar	100
int_crime	All Crime	Long Integer	30
int_social	Anti Social Behavior	Long Integer	30
int_burglary	Burglary	Long Integer	30
int_criminal_dam	Criminal damage and arson	Integer	20
int_drugs	Drugs	Integer	20
int_other_theft	Other theft	Integer	20
int_public_disorder	Public disorder and weapons	Integer	20
int_robbery	Robbery	Integer	20
int_shoplifting	Shoplifting	Integer	20
int_vehicle_crime	Vehicle crime	Integer	20
int_violent_crime	Violent crime	Integer	20
int_other_crime	Other crime	Integer	20
int_population	Population	Integer	15
int_vic_age	avg victim age	Integer	10
int_peak_time	Peak time	Date/Time	15
str_cities	uk Cities	Text/Varchar	10

1.3.1 Extracting of data into a temp table?

The screenshot displays the Oracle APEX SQL Workshop interface. The top navigation bar includes 'APEX', 'App Builder', 'SQL Workshop', 'Team Development', and 'Gallery'. The 'SQL Commands' tab is active, showing a SQL command in the editor:

```

1 select * from crime_fact
2 insert into crime_fact(crime_report_id, no_of_crime)
3 values(1, 24);
4

```

Below the editor, the 'Results' tab is selected, displaying the 'Index Columns' table:

Owner	Table Name	Index Name	Used in Plan	Columns	Uniqueness	Status	Index Type	Join Index
C7263731	CRIME_FACT	PK_CRIME_FACT		CRIME_REPORT_ID	UNIQUE	VALID	NORMAL	NO

The bottom of the interface shows the 'Query Plan' section and the system status bar, which includes the copyright notice 'Copyright © 1999, 2021, Oracle and/or its affiliates.' and the application version 'Application Express 21.2'.

OR

The screenshot shows the APEX SQL Workshop interface. The top navigation bar includes 'APEX', 'App Builder', 'SQL Workshop', 'Team Development', and 'Gallery'. The 'SQL Commands' tab is active, displaying a query editor with the following SQL code:

```
1 select * from crime_fact
2 join crime_type_dim;
```

Below the editor, there are tabs for 'Results', 'Explain', 'Describe', 'Saved SQL', and 'History'. The 'Saved SQL' tab is selected, showing a table with columns: Owner, Name, Description, SQL, Updated By, and Updated. The table is empty, with the message 'No Saved SQL Found' displayed below it.

Schema comparison

The screenshot shows the APEX Schema Comparison utility. The top navigation bar includes 'APEX', 'App Builder', 'SQL Workshop', 'Team Development', and 'Gallery'. The 'Utilities \ Schema Comparison' tab is active. The interface shows two schemas being compared: 'Schema 1' (C7263731) and 'Schema 2' (C7263731). The 'Compare' button is highlighted. Below the comparison settings, there is a table showing the results of the comparison.

Object Type	Object Name	Schema 1	Schema 2
SEQUENCE	CRIME_REGISTER_SEQ	✓	✓
SEQUENCE	FINANCIAL_SEQ	✓	✓
SEQUENCE	LOCATION_SEQ	✓	✓
SEQUENCE	OFFENCE_SEQ	✓	✓
SEQUENCE	OFFICER_SEQ	✓	✓
SEQUENCE	REGION_SEQ	✓	✓
TABLE	CRIME_BURGLRY	✓	✓
TABLE	CRIME_FACT	✓	✓
TABLE	CRIME_REGISTER	✓	✓
TABLE	CRIME_REPORTER	✓	✓

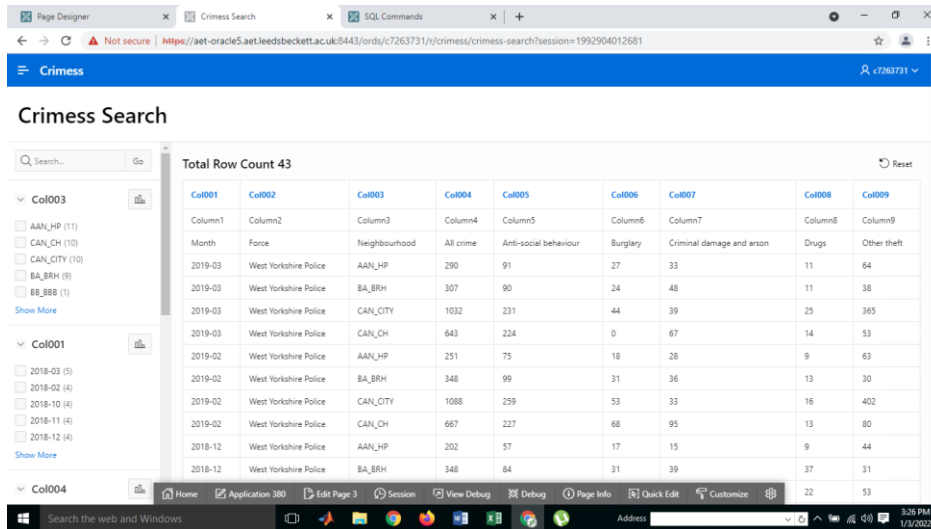
Cleaning of data

Answer: the crime dataset has been provided, there is no missing value and missing attribute.

Data is presented in data dictionary table followed by the data types and data item of interest.

Newly created table with dataset including all entries with no missing and no error appear in the dataset uploading and views.

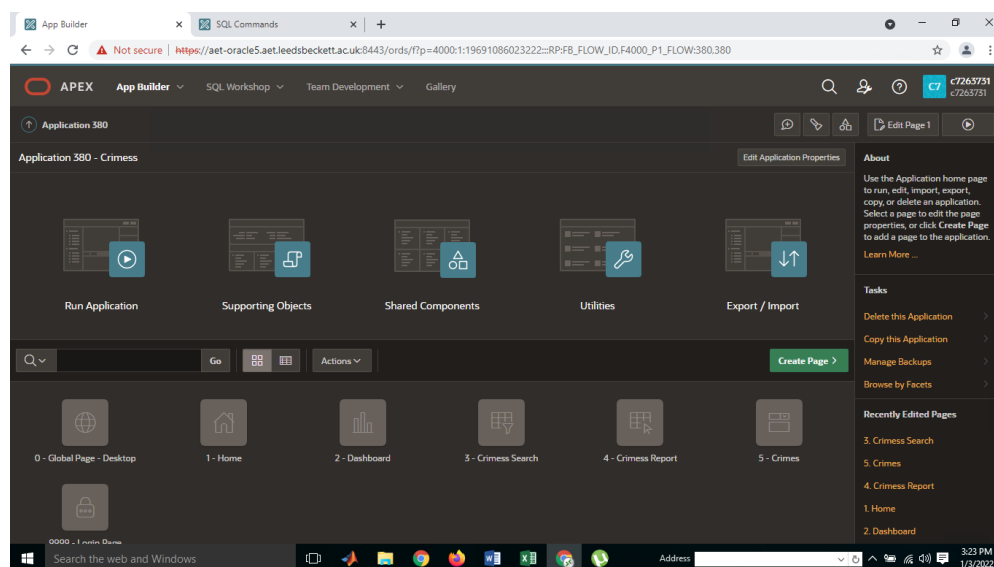
Crime dataset table followed by dataset



The screenshot shows the 'Crimess Search' application. On the left, there is a sidebar with filters for 'Col003' and 'Col001'. The main area displays a table with 43 rows and 9 columns. The columns are labeled Col001 through Col009. The data includes dates, police forces, neighborhoods, and various crime categories and counts.

Col001	Col002	Col003	Col004	Col005	Col006	Col007	Col008	Col009
Column1	Column2	Column3	Column4	Column5	Column6	Column7	Column8	Column9
Month	Force	Neighbourhood	All crime	Anti-social behaviour	Burglary	Criminal damage and arson	Drugs	Other theft
2019-03	West Yorkshire Police	AAN_HP	290	91	27	33	11	64
2019-03	West Yorkshire Police	BA_BRH	307	90	24	48	11	38
2019-03	West Yorkshire Police	CAN_CITY	1032	231	44	39	25	365
2019-03	West Yorkshire Police	CAN_CH	643	224	0	67	14	53
2019-02	West Yorkshire Police	AAN_HP	251	75	18	28	9	63
2019-02	West Yorkshire Police	BA_BRH	348	99	31	36	13	30
2019-02	West Yorkshire Police	CAN_CITY	1088	259	53	33	16	402
2019-02	West Yorkshire Police	CAN_CH	667	227	68	95	13	80
2018-12	West Yorkshire Police	AAN_HP	202	57	17	15	9	44
2018-12	West Yorkshire Police	BA_BRH	348	84	31	39	37	31

Application for checking & verifying dataset



Transforming of data

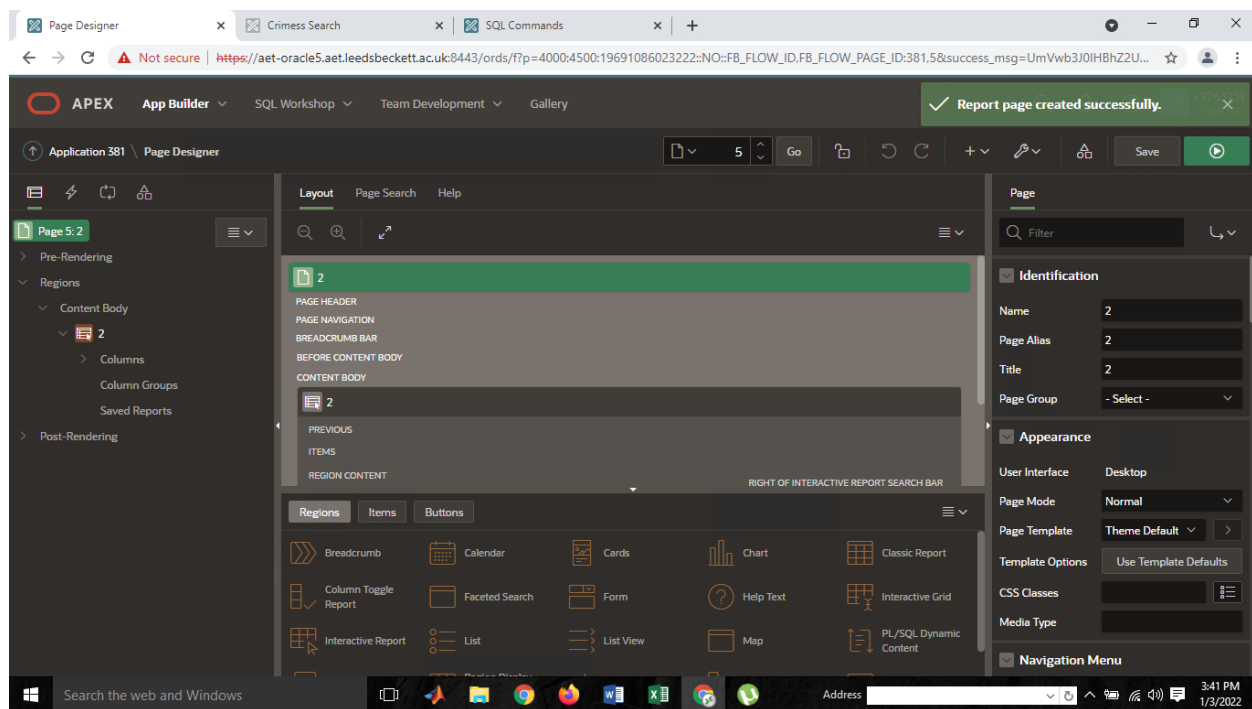
- **Design this ETL script. Have all considerations in a table.**

Answer:

(Rahm, E. Do, H.H., 2000) classify the quality of data, for cleaning the dataset is always considering a challenging part, in which various issues and problem has been addressed. Data cleaning when required when there is large amount of heterogeneous data is available. So there is need to improve the dataset efficiently and removing the unwanted rows and unwanted columns

In data cleaning removing the NA and NAN values in the dataset, in data warehousing dataset need to be cleared and precise , repeating values removed and only manage the data attributes with original value.

ETL SCRIPT IN APEX



Evidence code

The screenshot shows the APEX Page Designer interface. The left sidebar displays the page structure for 'Page 5:2', including regions and columns. The 'Columns' section is expanded, showing a list of columns: CRIME_REPORT_ID, FK1_S_ID, FK2_CT_ID, FK3_T_ID, NO_OF_CRIMES, and NO_OF_CT. The 'CRIME_REPORT_ID' column is selected. The main canvas shows a preview of the page layout with a 'Text [Hot]' component. The right sidebar shows the configuration for the selected column, including Identification, Heading, Single Row View, and Layout properties.

Page Designer

Application 381 | Page Designer

Layout | Page Search | Help

Column

CRIME_REPORT_ID

Type: Plain Text

Heading: Crime Report Id

Alignment: Left

Single Row View: Use Column Heading (checked)

Layout: Sequence 1, Column Alignment Left, Group - Select -

Loading data into SS

The screenshot shows the APEX SQL Workshop interface. The left sidebar displays the Object Browser with a list of tables. The 'CRIME_FACT' table is selected. The main canvas shows the table structure with columns: CRIME_REPORT_ID, FK1_S_ID, FK2_CT_ID, FK3_T_ID, NO_OF_CRIMES, and NO_OF_CT. The table is configured with a Primary Key on CRIME_REPORT_ID.

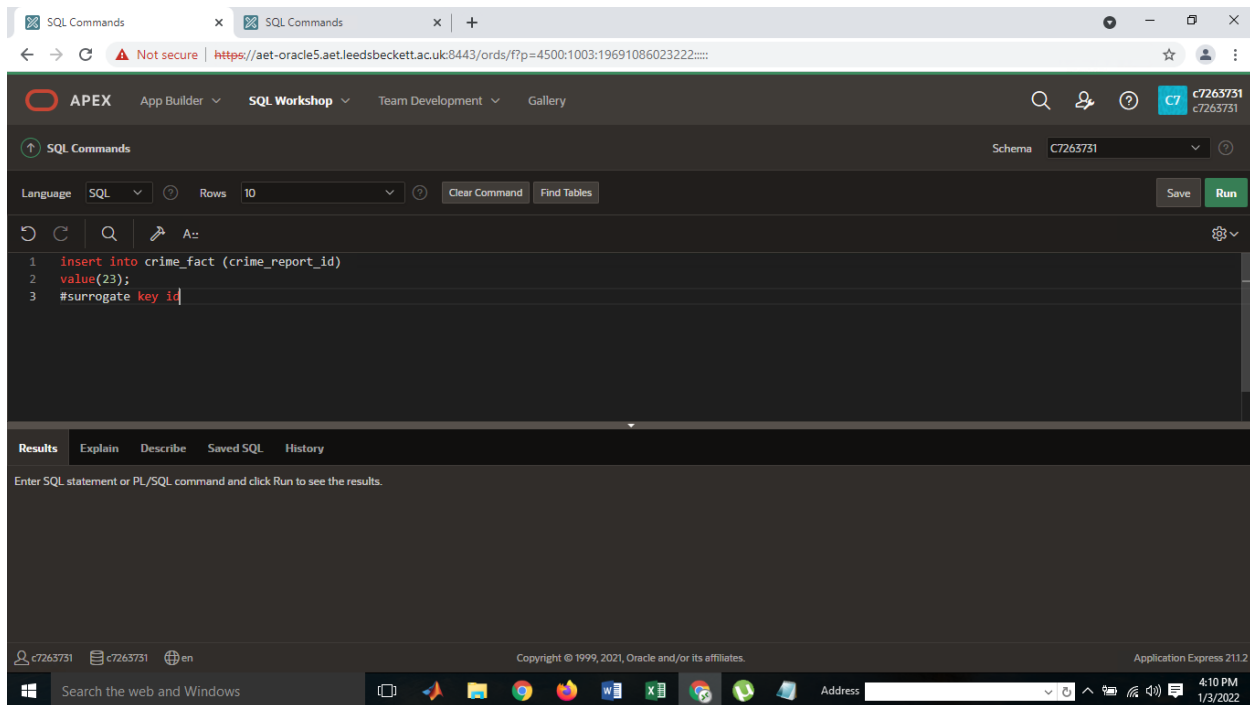
Object Browser

Schema: C7265731

CRIME_FACT

Column Name	Data Type	Nullable	Default	Primary Key
CRIME_REPORT_ID	NUMBER	No	-	1
FK1_S_ID	NUMBER	No	-	-
FK2_CT_ID	NUMBER	No	-	-
FK3_T_ID	NUMBER	No	-	-
NO_OF_CRIMES	NUMBER	No	-	-
NO_OF_CT	NUMBER	Yes	-	-

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Assumptions & Discussion

(Inmon, W.H., 1995) data warehousing was consider subject oriented and goal oriented approach to update & verify the time variant in the data dimensional table, most the data dictionary helps to generate the fact table and other associated data relational data in the database system. Data warehousing is large database processing unit system that extract, transform and load data attribute form different data sources based on the fact table of the organization. The relational schema of database works efficiently to manage and optimize the complex query operation on different source and destination locations.

Model of Fact Table

The screenshot shows the Oracle APEX SQL Workshop interface. The left pane displays the Object Browser with a list of tables, including CRIME_FACT. The main pane shows the Model view for the CRIME_FACT table. The model diagram illustrates a star schema with three dimension tables (TIME_DIM, CRIME_TYPE_DIM, STATION_DIM) connected to the fact table CRIME_FACT. The bottom status bar indicates the application is running on Oracle Express 211.2.

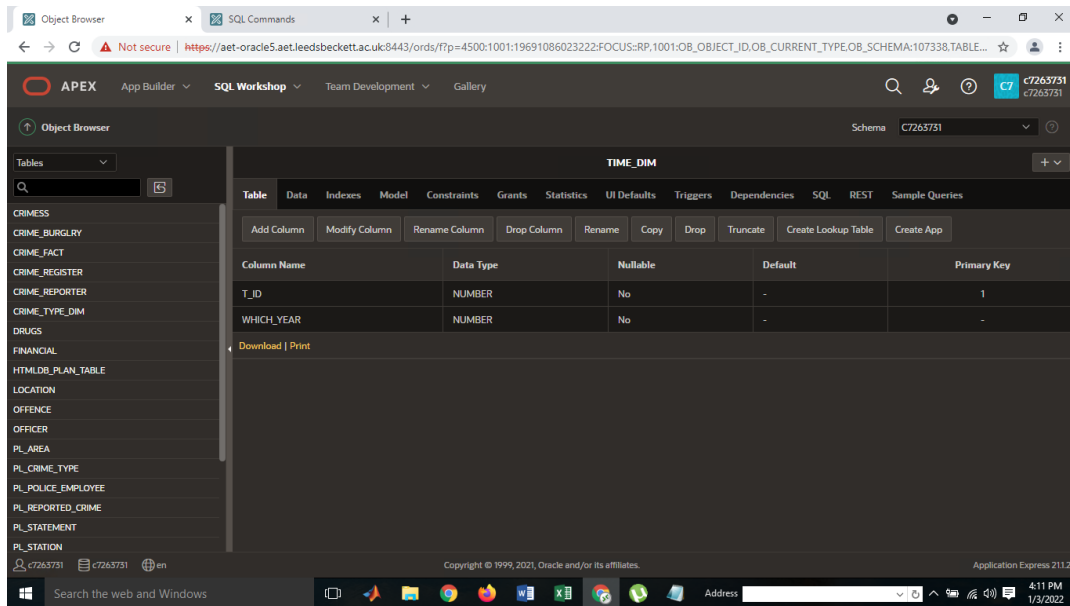
Fact table

The screenshot shows the Oracle APEX SQL Workshop interface. The left pane displays the Object Browser with a list of tables, including CRIME_FACT. The main pane shows the Table view for the CRIME_FACT table. The table structure is displayed with the following columns:

Column Name	Data Type	Nullable	Default	Primary Key
CRIME_REPORT_ID	NUMBER	No	-	1
FK1_S_ID	NUMBER	No	-	-
FK2_CT_ID	NUMBER	No	-	-
FK3_T_ID	NUMBER	No	-	-
NO_OF_CRIMES	NUMBER	No	-	-
NO_OF_CT	NUMBER	Yes	-	-

The bottom status bar indicates the application is running on Oracle Express 211.2.

How will you populate your Time Dim?



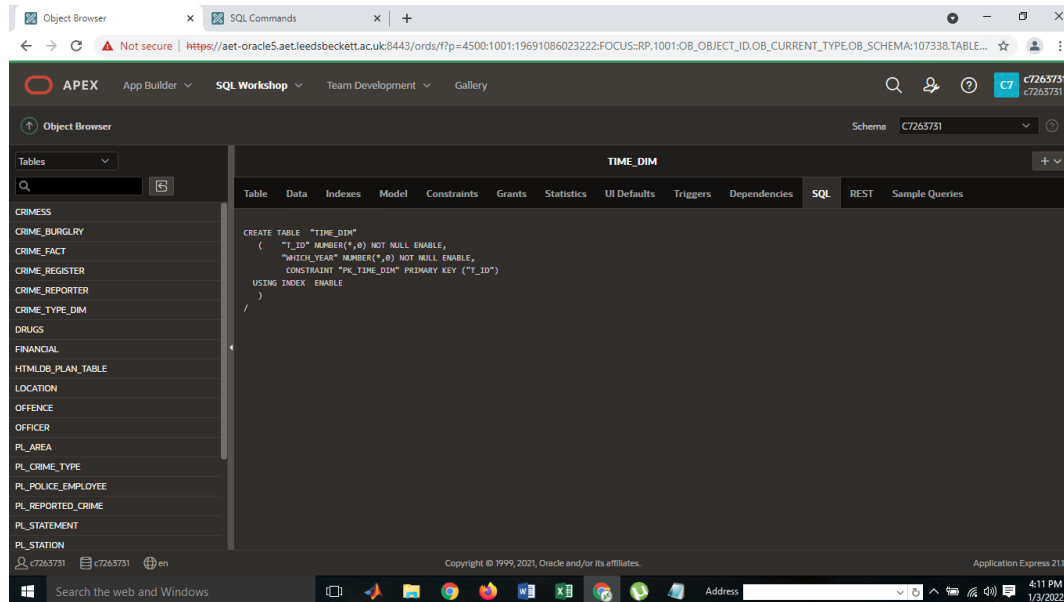
Evidence Code

```
CREATE OR REPLACE EDITIONABLE TRIGGER "CRIME_REGISTER_TRIG"
BEFORE INSERT ON CRIME_REGISTER
FOR EACH ROW
BEGIN
    IF :NEW.CRIME_id IS NULL THEN
        SELECT CRIME_REGISTER_SEQ.NEXTVAL INTO :NEW.CRIME_ID FROM
SYS.DUAL;
    END IF;
END;

/

ALTER TRIGGER "CRIME_REGISTER_TRIG" ENABLE
/ CREATE TABLE "CRIME_TYPE_DIM"
(
    "CT_ID" NUMBER(*,0) NOT NULL ENABLE,
    "C_TYPE" NUMBER(*,0) NOT NULL ENABLE,
    CONSTRAINT "PK_CRIME_TYPE_DIM" PRIMARY KEY ("CT_ID")
    USING INDEX ENABLE
)
/ CREATE TABLE "CRIME_BURGLRY"
(
    "BURGLR_ID" NUMBER(1,0),
    "BURGALRY_TOTAL" NUMBER(20,0),
    "FORCE_" VARCHAR2(20),
    "NEIGHBOUR" VARCHAR2(20),
    PRIMARY KEY ("BURGLR_ID")
```

USING INDEX ENABLE
)



Discussion

(Golab, L., Johnson, T., Seidel, 2009,) data warehousing generates the ETL script based statement view in order to process the query operation into the data warehousing projection. Data stream management system operate and verify the objects of internal fact table attribute. The similar data stream management system operate the associated table entries and operate the internal operation to check and verify the data integration operation. The real time data stream operation updated and verified on queries processing operating and on demand queries has been operated on request stream object. In data warehousing projections various largest queries operation has been processed and operate on data warehousing projection (Eckerson, W.W., 2002.) data quality depends on the data types and attributes association with the fact table, each id is connected with fact table to produce the associated query operation. The contribution of this report is going to investigate and interpret the resultant values based on the user choice and needs. The user query choice operation has been extracted load and integrate on the data dependent value.

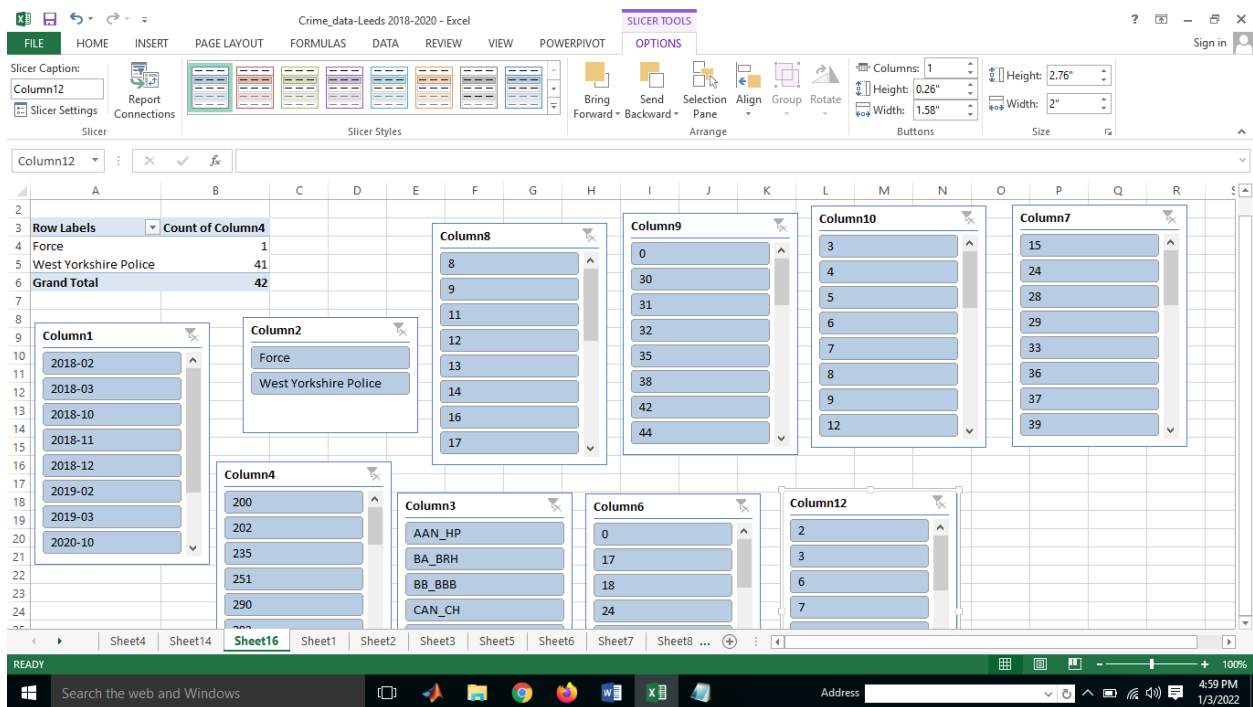
1.3.5 Ongoing Data Load (Data Maintenance (SCD)) implemented and evidenced as part to the solution.

2. TASK 2 - OLAP

PIVOT Table & Data Warehousing Query

(Gao, B., Zhang, Yao, N., 2012) in data warehousing mostly the data mining algorithm works to refine and extract the exact query required by the client and user of the organization, in data extraction and loading method the pivot table helps lot to generate and extract the query as per the on demand and requested by the client. The multi-dimensional model of data warehousing, data warehousing and OLAP provides the powerful method to process the complex query operated by the various source and destinations locations. The effective means of data dimensional object test and train the excel dataset and validate the result based on the testing training purposes. Pivot table approach previously applied by various data center to process and manage the query operation. In web based query extraction feature has been very poor and not good because of query processing time takes time while operate the query online method.

Excel Pivot Table



Fact Table

Attribute Name	Data type	Size	Foreign key (Dimension table)	Data type	size
CriminalReportID (PK)	Integer	10	CriminalReportID (PK)	Integer	10
S_id	Integer	10	S_id (FK)	Integer	10
T_ID	Integer	10	T_ID(FK)	Integer	10
CT_ID	Integer	10	CT_ID(FK)	Integer	10
NO_OF_CRIMES	Integer	10	NO_OF_CRIMES(FK)	Integer	10
NO_Of_CT	Integer	10	NO_Of_CT(FK)	Integer	10
			All_Criminal_ID (FK)	Integer	10

Dimension Table

Crime TYPE DIM

Attribute Name	Data type	Size
C_ID	Integer	10

TIME_DIM

Attribute Name	Data type	Size
T_ID	Integer	10

STATION_DIM

Attribute Name	Data type	Size
T_ID	Integer	10

OLAP Operation

The screenshot shows the Oracle APEX Object Browser interface. The left sidebar lists various objects in the CRIMESS schema, including CRIME_BURGLARY, CRIME_FACT, CRIME_REGISTER, CRIME_REPORTER, CRIME_TYPE_DIM, DRUGS, FINANCIAL, HTMLDB_PLAN_TABLE, LOCATION, OFFENCE, OFFICER, PL_AREA, PL_CRIME_TYPE, PL_POLICE_EMPLOYEE, PL_REPORTED_CRIME, PL_STATEMENT, and PL_STATION. The main pane displays the structure of the CRIMESS schema, showing a table with columns: ID, COL001, COL002, COL003, COL004, COL005, COL006, COL007, COL008, and COL009. The table has a primary key on the ID column. The interface includes tabs for Table, Data, Indexes, Model, Constraints, Grants, Statistics, UI Defaults, Triggers, Dependencies, SQL, REST, and Sample Queries. The bottom status bar shows the user is logged in as c7263731 and the application is running on Application Express 211.2.

OLAP Processing by describing

The screenshot shows the Oracle APEX SQL Commands window. The command entered is `1 describe time_dim`. The results are displayed in a table with columns: Table, Column, Data Type, Length, Precision, Scale, Primary Key, Nullable, Default, and Comment. The results show two columns for the TIME_DIM table: T_ID and WHICH_YEAR.

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
TIME_DIM	T_ID	NUMBER	22	-	0	1	-	-	-
	WHICH_YEAR	NUMBER	22	-	0	-	-	-	-

PIVOT Table in Excel

The screenshot shows Microsoft Excel with a PivotTable named 'PivotTable2' on the 'Sheet13' tab. The PivotTable has 'Column2' as the Row Labels and 'Count of Column4' as the Values. The data is as follows:

Row Labels	Count of Column4
Force	1
West Yorkshire Police	41
Grand Total	42

The 'PivotTable Fields' task pane on the right shows 'Column2' in the ROWS area and 'Count of Column4' in the VALUES area. The status bar at the bottom indicates the date is 1/3/2022.

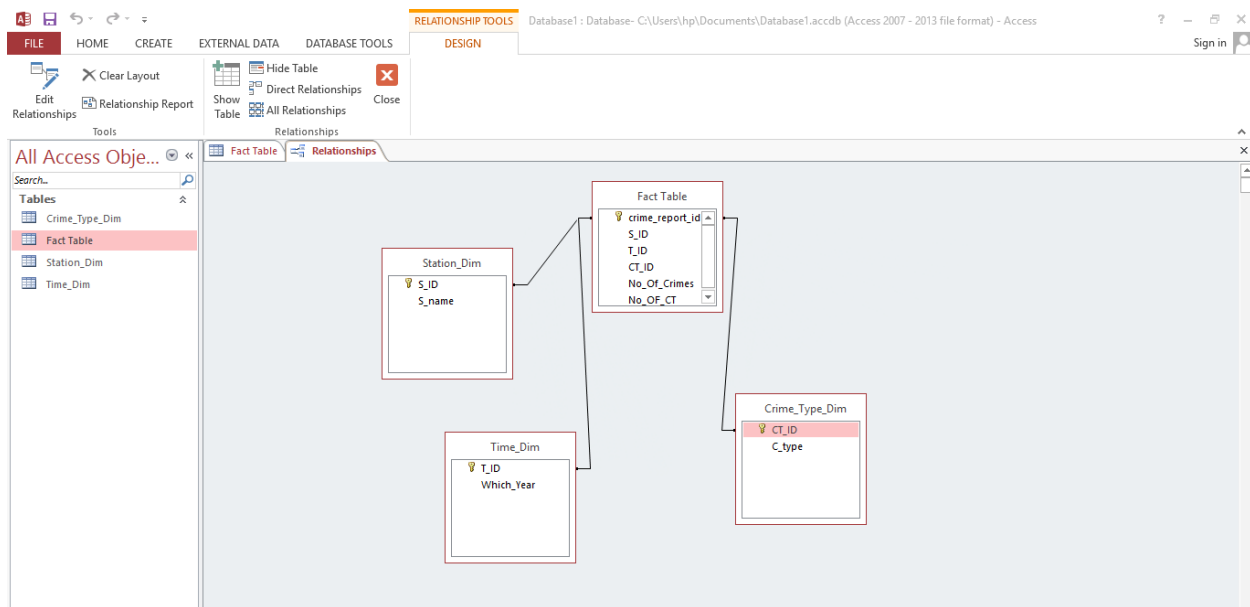
Database Tables in MS Access

The screenshot shows Microsoft Access in Design view for a table named 'Fact Table'. The table has the following fields:

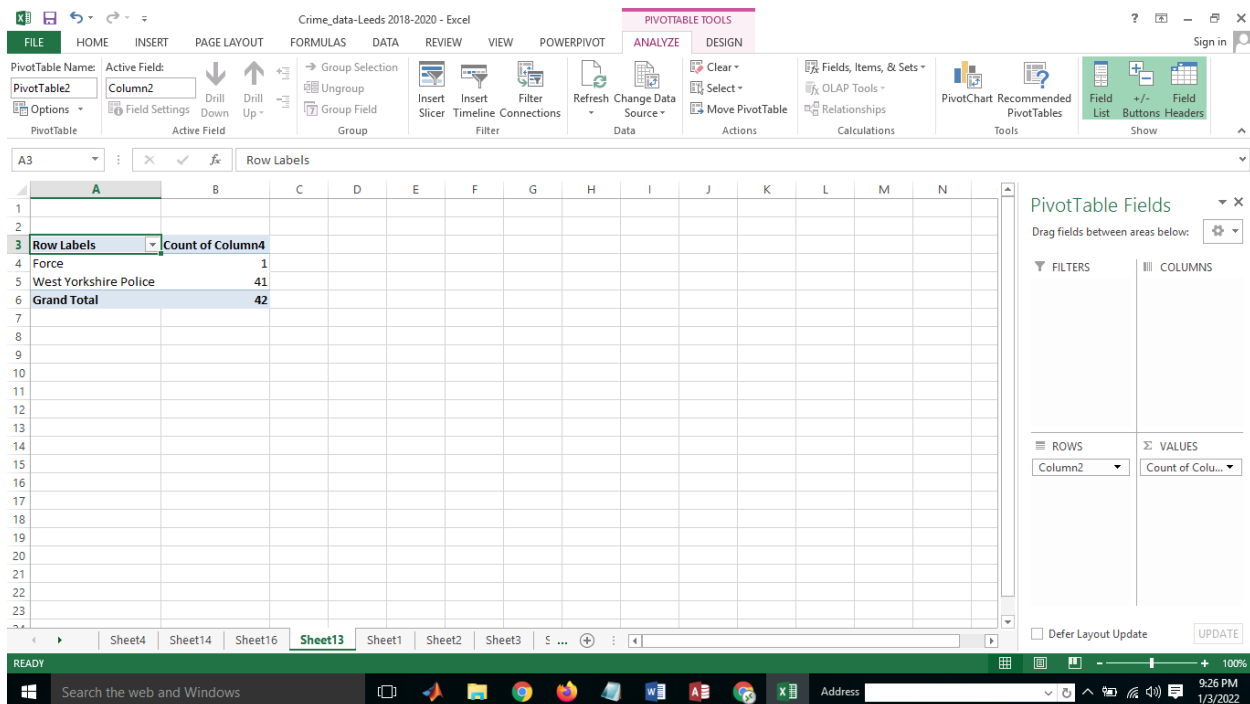
Field Name	Data Type	Description (Optional)
crime_report_id	AutoNumber	
S_ID	Number	
T_ID	Number	
CT_ID	Number	
No_Of_Crimes	Number	
No_Of_CT	Number	

The 'Field Properties' pane at the bottom shows the 'General' tab. A note at the bottom right states: 'A field name can be up to 64 characters long, including spaces. Press F1 for help on field names.'

Relationship mapping in Ms Access



PIVOT in EXCEL



Literature Review

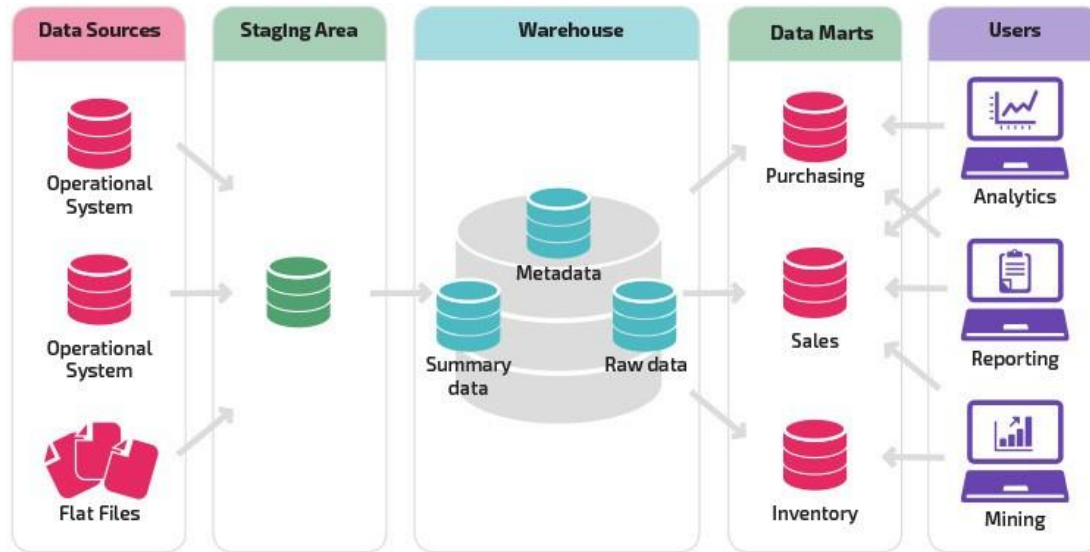
(Chenoweth, T., Corral, K 2006) data warehousing projection based on the data mart and its related technology to process the complex data warehousing query operation which has been analyzed and measured based on OLAP query structure and it process the largest operation to analyzed and measured the query. (Rahman, N., 2017) effective data ware housing approach is based on online data analytics tools & technique, which has been measured and predicted on query processing framework. The enterprise data warehousing projection based data mart which process the complex query operation that required by the client. The organization makes strategic business decision based on the data warehousing decision which extract and processed by the data mart. Motivation behind this studies scheme practically analyzed on ORACLE APEX tool various SQL query has been performed based on different database operation, data dimensional data has been created which associated with fact table of star schema design. In this studies fact table and dimensional data variable has been used to analyzed and process different OLAP query operation to take business and strategic decision to fulfill the business needs. TASK 3 – Findings and Reflection

Methodology

(Sen, A. Sinha, A.P., 2005) data warehousing methods (Aslam, T., 2020) data warehousing plays essential role for developing the organization business policies and boosting the business and employees growth level high due to online data analytics and machine learning method used by the data warehousing. Machine learning methods of logistic regression, naïve bays and support vector machine algorithm works effectively to mine the large dataset chunk and cubes and extract and analyzed the required data framework which required by the organization system. The development of data warehousing plays significant roles to coordinate time variation, nonvolatile assortment, and various data warehousing projections works online processing system. Client interface based on OLAP system, client can see related data cells by using the OLAP devices. Social database processed by the structured query language SQL operation. (Gardner, S.R., 1998) the data warehousing system design with the following steps:

1. Determine business objective
2. Collect and analyse information
3. Identify core business process

Architecture



PAPA framework, Theory of Stakeholders

(Bole, U., Popovič, A Jaklič, J., 2015.) Data warehousing framework based on data mining and data mart technique, data mining approach design on mining algorithm for decision making strategies to process the large amount of data. Data mining approach considering best approach for data warehousing projection in which particular data and information presented from the large pool of data warehouse. (Ciampi, F., Demi, S., Papa, A., 2021) data mining operate and processed the largest query operation based on innovative business model, the relationship between novel data mining approach which based on business model.

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Bole, U., Popovič, A., Žabkar, J., Papa, G. and Jaklič, J., 2015. A case analysis of embryonic data mining success. *International journal of information management*, 35(2), pp.253-259.

Ciampi, F., Demi, S., Magrini, A., Marzi, G. and Papa, A., 2021. Exploring the impact of big data analytics capabilities on business model innovation: The mediating role of entrepreneurial orientation. *Journal of Business Research*, 123, pp.1-13

Appendixes – add of your APEX code here

SQL Query Scripts

```
CREATE TABLE "TIME_DIM"
(
  "T_ID" NUMBER(*,0) NOT NULL ENABLE,
  "WHICH_YEAR" NUMBER(*,0) NOT NULL ENABLE,
  CONSTRAINT "PK_TIME_DIM" PRIMARY KEY ("T_ID")
  USING INDEX  ENABLE
)
CREATE TABLE "DRUGS"
(
  "DRUGS_ID" NUMBER(1,0),
  "DRUGS_TOTAL" NUMBER(20,0),
  "FORCE_" VARCHAR2(20),
  "NEIGHBOUR" VARCHAR2(20),
  PRIMARY KEY ("DRUGS_ID")
  USING INDEX  ENABLE
)
CREATE TABLE "FINANCIAL"
(
  "FINANCIAL_ID" NUMBER(8,0),
  "FINANCIAL_YEAR" NUMBER(4,0),
  "TOTAL_WALES_PER_CRIME" NUMBER(8,0) DEFAULT (0),
  CONSTRAINT "PK_FINANCIAL_ID" PRIMARY KEY ("FINANCIAL_ID")
  USING INDEX  ENABLE
)
/

CREATE OR REPLACE EDITIONABLE TRIGGER "FINANCIAL_TRIG"
BEFORE INSERT ON FINANCIAL
FOR EACH ROW
BEGIN
  IF :NEW.FINANCIAL_id IS NULL THEN
    SELECT FINANCIAL_SEQ.NEXTVAL INTO :NEW.FINANCIAL_ID FROM
SYS.DUAL;
  END IF;
END;

/
ALTER TRIGGER "FINANCIAL_TRIG" ENABLE

CREATE TABLE "CRIME_REPORTER"
(
  "REPORTER_ID" NUMBER,
  "REPORTER_NAME" VARCHAR2(35),
  "ADDRESS" VARCHAR2(60),
  PRIMARY KEY ("REPORTER_ID")
  USING INDEX  ENABLE
)
```

```

CREATE TABLE "CRIME_REGISTER"
(
  "CRIME_ID" NUMBER(4,0),
  "CRIME_NAME" VARCHAR2(20),
  "LOCATION_ID" NUMBER(8,0),
  "REPORTED_DATE" DATE NOT NULL ENABLE,
  "POLICE_ID" NUMBER(4,0),
  "CRIME_STATUS" VARCHAR2(8),
  "CLOSED_DATE" DATE DEFAULT NULL,
  "REPORTER_ID" NUMBER(8,0),
  "OFFICER_NOTE" VARCHAR2(128),
  "CRIME_TYPE" VARCHAR2(15),
  "LINKED_CRIME_ID" NUMBER(8,0),
  "DETECTIVE_ID" NUMBER(8,0),
  CHECK (crime_status in('OPEN', 'CLOSED', 'open','closed')) ENABLE,
  CONSTRAINT "PK_CRIME_REG" PRIMARY KEY ("CRIME_ID")
  USING INDEX ENABLE
)
/
ALTER TABLE "CRIME_REGISTER" ADD CONSTRAINT "FK_DETECTIVE" FOREIGN
KEY ("DETECTIVE_ID")
  REFERENCES "OFFICER" ("OFFICER_ID") ON DELETE CASCADE ENABLE
/
ALTER TABLE "CRIME_REGISTER" ADD CONSTRAINT "FK_CRIME_REG_LOC"
FOREIGN KEY ("LOCATION_ID")
  REFERENCES "LOCATION" ("LOCATION_ID") ON DELETE CASCADE
ENABLE
/
ALTER TABLE "CRIME_REGISTER" ADD CONSTRAINT "FK_OFFICER" FOREIGN
KEY ("POLICE_ID")
  REFERENCES "OFFICER" ("OFFICER_ID") ON DELETE CASCADE ENABLE
/

CREATE OR REPLACE EDITIONABLE TRIGGER "CRIME_REGISTER_TRIG"
BEFORE INSERT ON CRIME_REGISTER
FOR EACH ROW
BEGIN
  IF :NEW.CRIME_id IS NULL THEN
    SELECT CRIME_REGISTER_SEQ.NEXTVAL INTO :NEW.CRIME_ID FROM
SYS.DUAL;
  END IF;
END;

/
ALTER TRIGGER "CRIME_REGISTER_TRIG" ENABLE
/ CREATE TABLE "CRIME_TYPE_DIM"
(
  "CT_ID" NUMBER(*,0) NOT NULL ENABLE,

```

```

        "C_TYPE" NUMBER(*,0) NOT NULL ENABLE,
        CONSTRAINT "PK_CRIME_TYPE_DIM" PRIMARY KEY ("CT_ID")
    USING INDEX ENABLE
)
/ CREATE TABLE "CRIME_BURGLRY"
(
    "BURGLR_ID" NUMBER(1,0),
    "BURGALRY_TOTAL" NUMBER(20,0),
    "FORCE_" VARCHAR2(20),
    "NEIGHBOUR" VARCHAR2(20),
    PRIMARY KEY ("BURGLR_ID")
    USING INDEX ENABLE
)
Describe burglary
Describe time_dim

```

```

select count(*)
from crime_register;

```

```

select sum(location_id)
from crime_register;

```

```

select avg(location_id)
from crime_register;

```

```

select max(location_id)
from crime_register;

```

```

select min(location_id)
from crime_register;

```